SecureSignal: Hierarchical Multi-Agent DDQN with LSTM-Defended Traffic Sensing

***Abstract*—Urban traffic congestion costs the global economy hundreds of billions of dollars annually, yet deployed signal control systems remain static and are increasingly exposed to cyberattacks that corrupt sensor readings before they reach any decision-making logic. Existing deep reinforcement learning approaches either fail to coordinate across large intersection networks or operate without any defense against sensor manipulation. This paper presents *SecureSignal*, a system that addresses both problems through a progressive three-phase hierarchy of Double Deep Q-Network (DDQN) agents augmented with a training-free dual-layer sensor security mechanism. The control pipeline scales from a single intersection to four agents on a 2×2 grid and ultimately to eight agents governed by a two-tier Supervisor hierarchy, achieving a 94.3% reward gain over fixed-time control (Phase 0), a 68% zero-shot transfer improvement and an 86.8% fine-tuning gain (Phase 1), and a 4.6% improvement over the decentralized eight-intersection baseline (Phase 2). The security layer pairs a rolling-window Z-score anomaly detector (K=20, θ_z=3σ) with a pre-trained LSTM predictor (MAE $= 0.133$ vehicles/lane) to neutralize False Data Injection attacks, recovering average network reward to $-1{,}521.5$ within 0.1% of the unattacked baseline of $-1{,}522.6$ without retraining any reinforcement learning component. These results establish a principled, scalable, and attack-resilient foundation for intelligent urban traffic management.**

***Index Terms*—Intelligent networked transportation systems, Adaptive Urban Traffic Signal Control, Multi-agent reinforcement learning, Cyber Physical Systems Security, False Data Injection Attack**

I. Introduction

In today's modern era traffic congestion continues to be one of the most substantial challenge facing urban cities, leading to significant environmental, economic and public impacts. According to the INRIX Global Traffic Scorecard [22], American drivers lose an average of 54 hours per year to congestion, costing the U.S. economy an estimated $87 billion annually. Beyond its economic impact, traffic congestion increases vehicular emissions, contributing to increased greenhouse gas emissions and exacerbating the urban heat island effect. Furthermore, traffic congestion also degrades quality of life by increasing travel time, stress level and delaying emergency response. These adverse effects largely stem from the mismatch between static traffic signal control policies and the inherently dynamic nature of urban traffic flow.

Conventional traffic signal control systems primarily rely on fixed-time control policies, in which green phase durations are pre-computed offline using analytical approaches such as Webster's formula [23]. While analytically grounded, this approach cannot adapt to unpredictable demand fluctuations, accidents, or the cascading effects of congestion propagating across a road network. As reviewed by Maarouf et al. [20], reinforcement learning has emerged as the dominant paradigm for adaptive signal control over the past decade, with multi-agent RL (MARL) methods now accounting for over 60% of published TSC studies. Recent advances in Artificial Intelligence, particularly Reinforcement Learning (RL), offer a promising alternative for intelligent traffic signal control [7], [9]. RL frames traffic signal control as a Markov Decision Process (MDP), allowing an agent to learn an adaptive policy directly from environmental feedback without an explicit system model. Multi-Agent Reinforcement Learning (MARL) extends reinforcement learning to networked traffic control by assigning a dedicated agent to each intersection. However, the simultaneous policy updates of neighboring agents introduce a non-stationary learning environment, making convergence more challenging [11].

Although RL-based intelligent traffic signal control has emerged as a promising approach for urban traffic management, existing studies primarily focus on traffic efficiency while often neglecting security vulnerabilities and resilience against cyber-physical attacks. Networked traffic sensors transmit queue data over potentially insecure channels to control software. An adversary who intercepts and modifies these readings can silently corrupt the state observations fed to RL agents, inducing catastrophically suboptimal decisions without any conventional fault being detectable [16], [17]. Recent work has demonstrated that DRL-based adaptive traffic controllers are particularly vulnerable to coordinated sensor falsification by colluding vehicles [5]. Here, in this work we have proposed a unified hierarchical multi-agent Double Deep Q-Learning Network (DDQN) with a plug-in sensor security layer for adaptive and secure traffic signal control. The major contributions of this work are as follows:

1) **Hierarchical Multi-Agent Control.** A scalable distributed reinforcement learning framework is developed to train decentralized DDQN agents for intelligent traffic signal control across large urban road networks. By leveraging cooperative reward shaping, the framework facilitates implicit coordination among geographically distributed agents through continuous coordination signals, enabling network-level optimization without runtime message exchange. This approach is inspired by recent grouped and hierarchical MARL methods [1], [3], [10] and extends them with a two-tier Supervisor architecture. This communication-efficient architecture reduces coordination overhead while preserving

decentralized decision making, making it well suited for intelligent networked and distributed transportation systems [12], [14].

2) **Efficient Policy Transfer Across Network Scales.** A partial weight transfer mechanism is proposed to facilitate knowledge transfer across progressively larger traffic networks by reusing learned intersection-level feature representations. Building on evidence that hierarchical MARL policy transfer accelerates convergence in large-scale traffic networks [4], [8], the proposed approach allows newly instantiated agents to inherit previously learned policies instead of training from scratch, significantly reducing computational training cost and improving the scalability of distributed traffic signal control.
3) **Plug-In Sensor Security for Adaptive Controllers.** A plug-and-play, training-free dual-layer defense architecture is proposed to neutralize False Data Injection (FDI) attacks targeting distributed traffic sensing infrastructure. Motivated by demonstrated vulnerabilities of DRL controllers to adversarial sensor manipulation [5], [16], the framework performs real-time validation and correction of sensor measurements at the network edge using LSTM-based anomaly correction [9], [18], enabling decentralized adaptive controllers to make reliable decisions under adversarial conditions. By preserving controller performance without explicit retraining or centralized coordination, the proposed architecture provides a scalable and communication-efficient security solution for intelligent networked transportation systems.

The remainder of this paper is organized as follows. Section II reviews related work. Section III formalizes the problem. Section IV describes the proposed methodology. Section V presents the experimental setup. Section VI reports results and analysis. Section VII concludes our work.

II. Related Work

Deep Reinforcement Learning (DRL) based traffic signal control has gained substantial research momentum in recent years [20]. Early work established that DQN agents trained in simulation converge to policies substantially superior to fixed-time control, and subsequent developments introduced cooperative, hierarchical, and transfer-capable frameworks [7], [10].

A. Deep Reinforcement Learning for Traffic Signal Control

Early DQN-based approaches demonstrated that agents trained in simulation converge to policies substantially superior to fixed-time control [7], [19].Subsequent work established richer state representations and pressure-based reward signals through methods such as IntelliLight and PressLight, as surveyed in [20], while methods including CoLight and MetaLight [20] introduced attention-based neighbor coordination and meta-learning for rapid adaptation to new traffic patterns, and spatio-temporal hypergraph representations have recently been proposed to capture higher-order inter-intersection dependencies [11]. Recent frameworks have moved toward grouped and hierarchical architectures: GPLight [1] clusters intersections by traffic similarity for scalable parameter sharing; SHLight [10] employs a manager-worker hierarchy with importance sampling to address non-stationarity; and SERL-H [14] separates fast intersection-level actuation from slower region-level coordination under a centralized training with decentralized execution (CTDE) paradigm, reducing average delay from 45 s to 29 s in a 100-intersection SUMO simulation. HFRL [12] extends this to federated settings by grouping intersections for local federated averaging, while DTLight [8] achieves sample-efficient policy transfer via offline-to-online learning, reporting up to 40.7% improvement over online RL baselines, and Wu et al. [13] integrate CNN, LSTM, and graph attention networks for spatio-temporal feature extraction in large-scale hierarchical MARL validated in SUMO. The hybrid CNN-LSTM PPO model of Faqir et al. [9] further demonstrates that integrating spatiotemporal prediction into MARL improves efficiency by up to 92% for medium-demand scenarios, and the TG-MADDPG framework [15] similarly combines traffic flow prediction with graph attention for anticipatory multi-agent coordination. The Double DQN formulation [24], which decouples action selection from evaluation to mitigate overestimation bias, is the direct foundation of the agent in this work.

B. Security in Intelligent Transportation and RL Systems

FDI attacks, originally characterized in power grid state estimation [26], transfer directly to networked traffic sensor infrastructure. Qu et al. [5] formalized a colluding-vehicle attack model against DRL-based adaptive controllers, demonstrating fundamental vulnerability to coordinated observation-layer falsification even when the adversary controls only a small fraction of vehicles. Eghtesad et al. [16] modeled FDI detection in vehicular routing as a zero-sum game, computing a Nash equilibrium via multi-agent RL to provide worst-case-bounded travel time guarantees. Chen et al. [17] further showed that concurrent adversarial MARL training produces detectors robust to previously unseen injection patterns. On the detection side, rolling-window Z-score methods adapt automatically to local sensor statistics without labeled training data [26], while CNN-LSTM architectures have achieved strong anomaly classification accuracy in cyber-physical settings [6], [18]. The review by Maarouf et al. [20] further confirms that security and resilience remain the primary open challenges in deployed RL-based traffic signal control systems. The application of a pre-trained LSTM as a real-time sensor correction mechanism within a hierarchical RL control loop is, to the best of the authors' knowledge, novel to this work.

III. Problem Formulation

A. Traffic Network and MDP Formulation

The urban traffic network is modeled as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of intersections with $|\mathcal{V}| = N$ and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of directed road segments. The experimental road network was designed using a hierarchical scaling

strategy to evaluate the performance of the proposed Intelligent Transportation System at different levels of complexity. Initially, a single-intersection topology ($N = 1$) was considered in Phase 1. Subsequently, the network was expanded to a 2×2 grid containing four intersections ($N = 4$) in Phase 2, and finally to a 4×2 grid comprising eight intersections ($N = 8$) in Phase 3. This progressive scaling facilitates the analysis of scalability and performance under increasingly complex traffic scenarios.

The signal control task for each intersection i is formulated as an MDP $\mathcal{M}_i = (\mathcal{S}_i, \mathcal{A}_i, \mathcal{P}_i, \mathcal{R}_i, \gamma)$. The transition dynamics $\mathcal{P}_i$ are unknown and are implicitly approximated through model-free DDQN interaction with the environment. The local state at decision step t is:

$$\mathbf{s}_i^t = \left[q_N^i,\ q_S^i,\ q_E^i,\ q_W^i,\ \phi^i,\ \tau^i\right] \in \mathbb{R}^6, \tag{1}$$

where $q_d^i = \min(Q_d^i, 20)$ is the capped halting-vehicle count on incoming edge d; $\phi^i \in \{0, 1\}$ is the current signal phase; and $\tau^i = \min(\Delta t^i, 60)$ is the time elapsed since the last phase change. In Phase 1 cooperative mode, neighbor queue totals are appended, yielding $\mathbf{s}_i^t \in \mathbb{R}^8$. In Phase 2, the Supervisor appends a scalar coordination signal $\sigma_i \in [-1, +1]$, forming the enhanced state $\mathbf{h}_i^t = [\mathbf{s}_i^t;\ \sigma_i^t] \in \mathbb{R}^7$.

The action space is binary, $\mathcal{A}_i = \{0, 1\}$, where action 0 maintains the current phase and action 1 requests a switch. A minimum green duration of 5 seconds is enforced. The instantaneous reward is:

$$r_i^t = -\sum_{d\in\{N,S,E,W\}} q_d^i \;-\; 0.5\,W^i \;-\; 10\cdot\mathbf{1}\left[a_i{=}1 \wedge \tau^i{<}5\right], \tag{2}$$

where the first term uses the capped queue observation $q_d^i = \min(Q_d^i, 20)$ consistent with the agent's state, $W^i = \sum_{v\in\mathcal{U}_i} w_v$ is cumulative per-vehicle waiting time summed over the set $\mathcal{U}_i$ of vehicles currently on incoming edges at intersection i, and the indicator term penalizes premature switching. Each agent minimizes the DDQN temporal difference loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathcal{D}}\left[\left(r + \gamma\, Q_{\theta^-}\left(s',\ \arg\max_{a'} Q_\theta(s', a')\right) - Q_\theta(s, a)\right)^2\right], \tag{3}$$

where θ and θ^- denote online and target network parameters, $\mathcal{D}$ is the experience replay buffer, and $\gamma = 0.95$.

B. Security Problem Formulation

An adversary is assumed to intercept and modify sensor state readings before they reach the RL agents, but cannot access agent parameters, alter the physical environment, or observe the reward signal. This threat model is consistent with demonstrated FDI vulnerabilities in networked traffic sensor infrastructure, where even a small fraction of compromised sensors can induce network-wide performance degradation [5], [16], [17]. At each step t, intersection i is targeted with probability $p_{\text{fdi}} = 0.15$. A single direction $d^* \in \{N, S, E, W\}$ is sampled uniformly and receives a perturbation $\delta \sim \text{Uniform}[10.0,\ 15.0]$ vehicles; only one direction per targeted intersection is attacked per step:

$$\tilde{q}_{d^*}^i = \text{clip}\left(q_{d^*}^i + \delta,\ 0,\ 20\right). \tag{4}$$

A complementary non-adversarial noise model drops packets with probability $p_{\text{loss}} = 0.05$ and delays survivors by up to 3 steps. The defender's objective is to detect anomalous readings and recover plausible state estimates without modifying any RL component.

TABLE I
MATHEMATICAL NOTATION

Symbol	Definition
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$	Traffic network graph
$N = \lvert\mathcal{V}\rvert$	Number of intersections
$\mathbf{s}_i^t \in \mathbb{R}^6$	Local state of intersection i at step t
$\mathbf{h}_i^t \in \mathbb{R}^7$	Enhanced state (local + supervisor signal)
q_d^i	Halting vehicle count, direction d, intersection i
$\phi^i \in \{0, 1\}$	Current signal phase
τ^i	Time since last phase change (seconds)
$a_i \in \{0, 1\}$	Agent action (0 = keep, 1 = switch)
r_i^t	Instantaneous reward for agent i at step t
W^i	Cumulative per-vehicle waiting time at intersection i
$\gamma = 0.95$	Discount factor
$\theta,\ \theta^-$	Online and target DDQN network parameters
$\mathcal{D}$	Experience replay buffer
$\sigma_i \in [-1, +1]$	Supervisor coordination signal for agent i
$\mathcal{U}_i$	Set of vehicles on incoming edges at intersection i
$\mathbf{g} \in \mathbb{R}^{24}$	Concatenated group state (4×6)
$\mathbf{m} \in \mathbb{R}^4$	Cross-group summary vector
$\psi,\ \psi^-$	Online and target Supervisor network parameters
r^{grp}	Group-average reward
$p_{\text{fdi}} = 0.15$	FDI attack probability per intersection/step
$\delta \in [10, 15]$	Injected queue perturbation (vehicles)
$H_{i,d}$	Rolling history buffer for direction d at intersection i
$\mu_{i,d},\ \sigma_{i,d}$	Mean and std. of $H_{i,d}$
$z_{i,d}$	Z-score of queue reading at (i, d)
$\theta_z = 3.0$	Z-score anomaly detection threshold
$K = 20$	Rolling history window size (steps)
$p_{\text{loss}} = 0.05$	Packet loss probability
ε	Numerical stability constant (10^{-6})

C. Notation Summary

Table I lists all mathematical symbols used in this paper. All reward figures in Section VI are presented on two scales: *network total* (sum across all N agents per episode) and *per-intersection average* (network total divided by N), distinguished explicitly throughout.

IV. Proposed Methodology

The end-to-end architecture of SecureSignal, illustrated in Fig. 1, comprises three functional layers: the SUMO traffic simulation environment, the dual-layer security preprocessing stage, and the hierarchical DDQN control layer. State signals flow upward from environment through security to control; actions flow downward from control to environment.

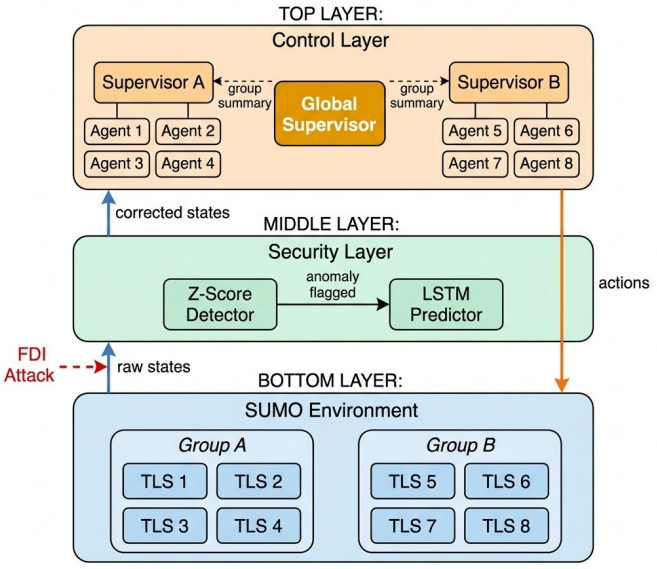


Fig. 1. End-to-end system architecture of SecureSignal. Three horizontal layers are shown: SUMO environment (bottom), security layer (middle), and hierarchical DDQN control layer (top). Arrows indicate signal flow; the red dashed arrow marks the FDI injection point.

Algorithm 1 Single-Intersection DDQN Training (Phase 0)

Require: SUMO env. $\mathcal{E}$; Q_θ, Q_{θ^-}; buffer $\mathcal{D}$; ε_0=1.0, $\varepsilon_{\min}$=0.01, ρ=0.995; α=0.001; B=64; C=10; N_{ep}=1000

Ensure: Trained checkpoint Q_θ^*

1: **for** episode $= 1$ **to** N_{ep} **do**
2: $s \leftarrow \mathcal{E}.\text{reset}()$
3: **while** not done **do**
4: $a \leftarrow \varepsilon\text{-greedy}(Q_\theta, s)$
5: $s', r, \text{done} \leftarrow \mathcal{E}.\text{step}(a)$; $\mathcal{D}.\text{store}(s, a, r, s', \text{done})$
6: **if** $|\mathcal{D}| \geq B$ **then**
7: $\mathcal{B} \sim \mathcal{D}$; $y \leftarrow r + \gamma\, Q_{\theta^-}(s', \arg\max_{a'} Q_\theta(s', a'))$
8: Adam [28] step on $\frac{1}{B}\sum(y - Q_\theta(s,a))^2$; clip grad to 10.0
9: **end if**
10: $s \leftarrow s'$
11: **end while**
12: $\varepsilon \leftarrow \max(\varepsilon_{\min}, \varepsilon \cdot \rho)$
13: **if** episode $\bmod C = 0$ **then**
$\theta^- \leftarrow \theta$
14: **end if**
15: **end for**
16: **return** Q_θ^*

A. Phase 0 — Single-Intersection DDQN Baseline

A single DDQN agent controlling one isolated intersection provides the foundational policy that subsequent phases inherit. The neural network is a three-layer MLP, $\text{Linear}(6, 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128, 128) \rightarrow \text{ReLU} \rightarrow \text{Linear}(128, 2)$, with Xavier-uniform weight initialization. Algorithm 1 formalizes the complete training procedure.

The Double DQN target (line 7) selects the next action using the online network but evaluates it using the target network, eliminating the overestimation bias of vanilla DQN [24]. The hard target update every 10 episodes provides a stable regression target, while gradient clipping to L2 norm 10.0 prevents divergence during early high-variance exploration. After 1,000 training episodes the agent achieves a 94.3% improvement in cumulative episode reward over the Webster fixed-time baseline [23], confirming that the learned policy exploits real-time queue asymmetries that static timing cannot capture. This result aligns with recent SUMO-based evaluations by Fereidooni et al. [7], who similarly reported that DRL policies significantly outperform Webster-formula baselines across diverse traffic scenarios. This figure is a reward-based metric; the associated waiting-time outcome is reported separately in Table VI.

B. Phase 1 — Cooperative DDQN Transfer and Fine-Tuning

The Phase 0 checkpoint is transferred to a 2×2 grid of four intersections (TLS_1 through TLS_4), providing a warm-start policy that bypasses initial random exploration. Two variants are evaluated. In *independent mode*, each agent retains the 6-dimensional state and individual reward; weights are cloned directly from Phase 0 and fine-tuned for 100 episodes with $\alpha = 0.0001$. In *cooperative mode*, the state is augmented to 8 dimensions by appending the total queue lengths at the two topologically adjacent neighbors, and a shared group-average reward $r_{\text{coop}} = \frac{1}{4}\sum_{k=1}^{4} r_k$ replaces individual rewards; because the increased state dimension prevents direct weight cloning, cooperative agents are initialized with Xavier weights and trained from scratch for 700 episodes with $\alpha = 0.0005$. The shared group-average reward is a sufficient coordination signal to achieve near-perfect load balancing: agents cannot improve their local metric by starving neighboring intersections of throughput, since all rewards are identical. This cooperative reward shaping strategy is consistent with the grouped-reward designs used in large-scale MARL traffic control [1], [2]. Zero-shot transfer of the Phase 0 checkpoint immediately achieves a network reward of $-1{,}363.1$ — a 68% improvement over a random-policy baseline — before any additional training, confirming that the local state features generalize meaningfully to the multi-intersection setting [8], [25]. Independent fine-tuning over just 100 episodes reaches a network reward of -560.8, an 86.8% improvement over the zero-shot baseline.

C. Phase 2 — Hierarchical DDQN with Supervisor Agents

Eight intersections are partitioned into Group A $= \{\text{TLS}_{1..4}\}$ and Group B $= \{\text{TLS}_{5..8}\}$. Each group is supervised by a SupervisorAgent: $\text{Linear}(d_{\text{in}}, 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64, 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64, 4) \rightarrow \text{Tanh}$, outputting four coordination signals $\boldsymbol{\sigma} \in [-1, +1]^4$. The Local Supervisor uses $d_{\text{in}} = 24$ (four agents × 6-dim states). The Global Supervisor uses $d_{\text{in}} = 28$, incorporating a 4-dim cross-group

Algorithm 2 Hierarchical DDQN Core Loop (Phase 2)

Require: Phase 1 checkpoint $\{Q^*_{\theta_i}\}$; Supervisors f_{ψ_A}, f_{ψ_B}; global flag; N_{ep} episodes
Ensure: $\{Q^*_{\theta_i}\}$, $f^*_{\psi_A}$, $f^*_{\psi_B}$

Copy cols 0:6 of W_1 to $W_2[:, 0{:}6]$; init $W_2[:, 6]$ Xavier {partial weight transfer}
for episode $= 1$ **to** N_{ep} **do**
  $\{s_i^{\text{loc}}\} \leftarrow \mathcal{E}.\text{reset}()$
  if global flag **then**
    $\boldsymbol{\sigma}_A \leftarrow f_{\psi_A}([\mathbf{g}_A; \mathbf{m}_B])$; $\boldsymbol{\sigma}_B \leftarrow f_{\psi_B}([\mathbf{g}_B; \mathbf{m}_A])$
  else
    $\boldsymbol{\sigma}_A \leftarrow f_{\psi_A}(\mathbf{g}_A)$; $\boldsymbol{\sigma}_B \leftarrow f_{\psi_B}(\mathbf{g}_B)$
  end if
  $h_i \leftarrow [s_i^{\text{loc}};\ \sigma_i]$ for all i
  while not done **do**
    $\{a_i\} \leftarrow \varepsilon$-greedy$(\{Q_{\theta_i}\}, \{h_i\})$; $\{s_i'^{\text{loc}}\}, \{r_i\} \leftarrow \mathcal{E}.\text{step}(\{a_i\})$
    Train $\{Q_{\theta_i}\}$ on r_i; train $f_{\psi_{A/B}}$ on $r^{\text{grp}}_{A/B}$ via Eq. 5
    Update $\{h_i\}$ from recomputed supervisor signals
  end while
  Decay ε; sync target networks every C episodes
end for
return $\{Q^*_{\theta_i}\}$, $f^*_{\psi_A}$, $f^*_{\psi_B}$

summary $\mathbf{m} = [\bar{q},\ q_{\max},\ \bar{w},\ q_{\text{boundary}}]$. Partial weight transfer maps the Phase 1 input weight matrix $W_1 \in \mathbb{R}^{128\times 8}$ to $W_2 \in \mathbb{R}^{128\times 7}$ by copying the six local-feature columns exactly and initializing the seventh (supervisor signal) column with Xavier initialization, preserving prior feature representations while enabling the agent to learn the new signal-to-action mapping from scratch. This partial transfer design directly extends the hierarchical MARL transfer methodology of Li et al. [4] and the offline-to-online transfer principles of Huang et al. [8], and the transferability analysis established by Yosinski et al. [25]. The two-tier supervisor hierarchy is further motivated by the manager-worker architectures of both Zhao et al. [3] and Shen [10], adapted here to emit continuous coordination signals rather than discrete sub-goals. The CTDE paradigm adopted in SERL-H [14] and HFRL [12] further validates the architectural choice of separating region-level coordination from intersection-level actuation. The Supervisor's TD loss is:

$$\mathcal{L}_{\text{sup}}(\psi) = \mathbb{E}_{\mathcal{D}^{\text{sup}}}\left[\left(r^{\text{grp}} + \gamma\,\overline{f_{\psi^-}(\mathbf{g}')} - f_\psi(\mathbf{g})\right)^2\right], \quad (5)$$

where the overline denotes the mean of the target Supervisor's output vector, broadcast as a scalar update across all four signal dimensions. Algorithm 2 summarizes the novel elements of the hierarchical training loop.

D. Security Layer — Dual-Layer Defense

To address the vulnerabilities of distributed cyber-physical systems, the SecurityLayer is designed as a stateful, edge-based preprocessing wrapper inserted between the networked SUMO sensor outputs and all downstream RL computation.

Algorithm 3 SecurityLayer.process (raw_states, t)

Require: raw_states $= \{\mathbf{s}_i^t\}$; operating mode m; step t
Ensure: Corrected output states $\{\hat{\mathbf{s}}_i^t\}$

states $\leftarrow$ deep copy of raw_states
if $m \in \{\texttt{attack}, \texttt{defense}, \texttt{secure}\}$ **then**
  states $\leftarrow$ APPLYFDI(states, t) {Eq. 4, prob. p_{fdi}}
end if
if $m \in \{\texttt{unreliable}, \texttt{secure}\}$ **then**
  states $\leftarrow$ APPLYNETWORKNOISE(states, t)
end if
if $m \in \{\texttt{defense}, \texttt{secure}\}$ **then**
  states $\leftarrow$ DETECTANDCORRECT(states, t)
end if
return states

Algorithm 4 DetectAndCorrect (states, t)

Require: states; buffers $\{H_{i,d}\}$; LSTM; K=20; θ_z=3.0; ε=10^{-6}
Ensure: Corrected states

for all intersection i **do**
  $\mathbf{q} \leftarrow \mathbf{s}_i^t[0{:}4]$
  if $|H_{i,\cdot}| < 10$ **then**
    Append $\mathbf{q}$; **continue** {warm-up}
  end if
  $z_{i,d} \leftarrow |q_d - \mu_{i,d}|/(\sigma_{i,d} + \varepsilon)$ for all d
  flagged $\leftarrow \{d : z_{i,d} > \theta_z\}$
  if flagged $\neq \emptyset$ **then**
    $\hat{\mathbf{q}} \leftarrow$ LSTM(last K entries of H_i)
    for all $d \in$ flagged **do**
      $q_d \leftarrow \text{clip}(\hat{q}_d,\ 0,\ 20)$
    end for
    Increment detection counter
  end if
  Append corrected $\mathbf{q}$ to $H_{i,d}$
end for
return corrected states

Raw local states $\{\mathbf{s}_i^t\}$ pass through Algorithm 3 at every decision step before being used by any supervisor or agent.

The Z-score threshold of 3σ is chosen to be sensitive to FDI injections of $[10, 15]$ vehicles — which produce Z-scores of $[15, 50]$ for typical clean queue distributions — while remaining robust to natural inter-episode variability. The LSTM correction is preferable to mean substitution because real traffic queues follow time-series trends whereas FDI spikes are instantaneous; an LSTM trained on 50 clean episodes captures these dynamics and enables corrections that track the true state rather than reverting to a stationary prior. This makes the architecture highly suitable for decentralized network computing environments where edge nodes must independently validate streaming sensor data. The integration of CNN-LSTM modules for anticipatory state estimation in networked environments [9], [15] directly motivates the use of a sequential LSTM predictor for sensor correction rather than

TABLE II
SIMULATION AND SECURITY LAYER PARAMETERS

Parameter	Value
Simulator	SUMO v1.25.0
Decision interval Δ	5 seconds
Phase 0/1 episode duration	3,600 s (720 steps)
Phase 2 / security episodes	1,800 s (360 steps)
Traffic generation	Stochastic (`-random`)
Min. green duration	5 seconds
Queue cap	20 vehicles/direction
Random seed	42
p_{fdi} / $[\delta_{\min}, \delta_{\max}]$	0.15 / [10, 15] veh.
θ_z / K / min. history	3.0 / 20 / 10 steps
p_{loss} / max. delay	0.05 / 3 steps
LSTM hidden / layers / epochs	64 / 1 / 25
LSTM lr / batch / grad clip	0.001 / 256 / 5.0

TABLE III
AGENT AND SUPERVISOR HYPERPARAMETERS ACROSS ALL PHASES

Parameter	Ph. 0	P1 Ind.	P1 Co.	Ph. 2	Loc.Sup	Glob.Sup
State dim.	6	6	8	7	24	28
Hidden size	128	128	128	128	64	64
Output dim.	2	2	2	2	4	4
Lr	0.001	0.0001	0.0005	0.0001	0.001	0.001
ε_0	1.0	0.1	0.9	0.3	–	–
Batch / Buf.	64/10k	64/10k	64/10k	64/10k	64/10k	64/10k
Grad clip	10.0	10.0	10.0	10.0	10.0	10.0
Episodes	1,000	100	700	500	500	900
Init.	Xavier	Transfer	Xavier	Partial	Xavier	Xavier

static imputation. LSTM-based anomaly detection and state reconstruction has demonstrated strong performance in cyber-physical systems under adversarial sensor manipulation [6], [18], [27], and the colluding-vehicle attack scenario studied by Qu et al. [5] and the zero-sum FDI game formulation of Eghtesad et al. [16] confirm that such manipulation is a realistic and strategically optimizable threat to deployed DRL traffic controllers. As quantified in Fig. 4, the LSTM reduces per-instance sensor error from 17.0 to 0.62 vehicles on a representative attack, a 96.4% reduction in corruption magnitude.

V. Experimental Setup

A. Simulation Environment and Baseline Configurations

All experiments use SUMO v1.25.0 via the TraCI Python API. The `-time-to-teleport -1` flag prevents vehicle teleportation; `-waiting-time-memory 10000` enables accurate per-vehicle waiting time tracking. Vehicle spawning is randomized per episode in Phase 2 and security evaluations. Table II lists key parameters.

Six configurations are evaluated. *Fixed-Time Control* uses Webster-formula [23] 30-second alternating green phases and serves as the Phase 0 benchmark. *Single-Intersection DDQN* is the fully trained Phase 0 agent. *Cooperative DDQN* comprises four agents with 8-dimensional states and shared group rewards trained for 700 episodes. *Hierarchical DDQN with Local Supervisor* deploys eight agents with 7-dimensional enhanced states from two 24-input Supervisors trained for 500 episodes. *Hierarchical DDQN with Global Supervisor* expands each Supervisor to 28-dimensional inputs with cross-group summaries, trained for 900 episodes. *Security Evaluation* applies all five scenario modes to the frozen Phase 2 Local Supervisor checkpoint, explicitly testing the system's resilience against both adversarial data injection and realistic network communication degradation (packet loss and latency).

B. Hyperparameters and Evaluation Metrics

Table III lists all agent and supervisor hyperparameters. Learning rates decrease progressively across phases (0.001 in Phase 0, 0.0001 in Phase 2), reflecting the decreasing need for aggressive updates as policies approach convergence. Phase 1 cooperative training uses an intermediate rate of 0.0005 to balance the larger state space against the warm-start initialization. The Supervisor's rate of 0.001 is higher than local agents because its rapidly changing group-state input requires faster adaptation.

Traffic performance is quantified by cumulative episode reward (network total), average reward per intersection (network total divided by N), and average per-vehicle waiting time in seconds. These two reward scales are distinguished explicitly throughout Section VI. For security experiments, two additional metrics are tracked: total detection events per episode (lanes flagged by the Z-score detector) and detection rate (detections normalized by attack activation events). The false positive rate is discussed in Section VII.

VI. Results and Discussion

Scale note. *Network total reward* is the sum across all N agents per episode. *Per-intersection reward* divides this by $N{=}8$ for Phase 2 and security, or $N{=}4$ for Phase 1. The abstract cites Phase 2 results at -187.9 per intersection; Table IV reports security on the network-total scale ($-1{,}522.6$). The conversion is $-1{,}522.6 \div 8 = -190.3$ per intersection; the slight discrepancy from -187.9 arises from the shorter 1,800 s episode duration used in security evaluation versus 3,600 s during training.

A. Phase 0 and Phase 1 Results

After 1,000 training episodes, the Phase 0 DDQN agent achieves a 94.3% improvement in cumulative episode reward over the Webster fixed-time baseline. The policy exploits real-time queue asymmetries — allocating green time to loaded approaches and suppressing unnecessary switches on lightly loaded phases — capabilities structurally absent in fixed-time control, consistent with recent SUMO-based DRL evaluations [7], [19]. This figure is a reward-based metric; the associated waiting-time reduction is a separately measured outcome reported in Table VI.

Zero-shot transfer of the Phase 0 checkpoint to the four-intersection grid immediately yields a network reward of $-1{,}363.1$, a 68% improvement over a random-policy baseline with no additional training, confirming that the learned queue-management representations generalize across intersections [25]. This zero-shot generalization mirrors the offline

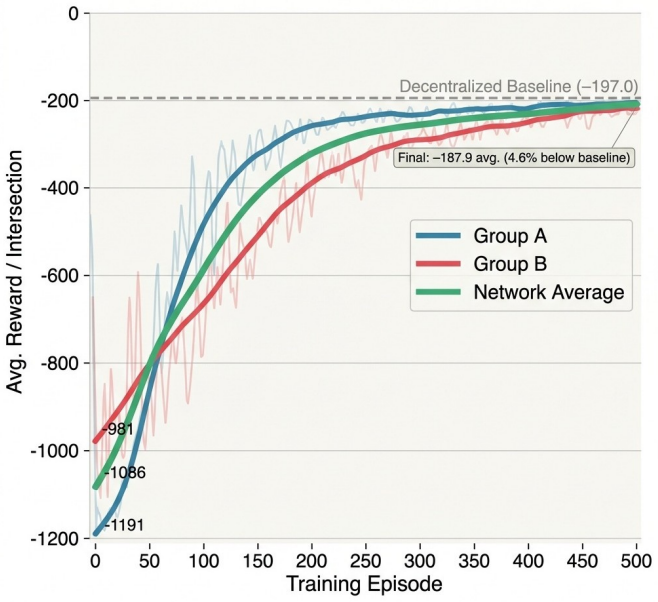


Fig. 2. Average reward per intersection across 500 Phase 2 training episodes for Group A, Group B, and the eight-intersection network average, with the decentralized baseline (-197.0) shown as a horizontal dashed reference line. Both group curves converge in parallel, demonstrating balanced cross-cluster coordination.

pretraining advantage reported in DTLight [8], where pre-collected policy knowledge was shown to transfer successfully across traffic scenarios without additional environment interaction. Independent fine-tuning over 100 episodes achieves -560.8, an 86.8% improvement over the zero-shot starting point, at a convergence speed that would require orders of magnitude more episodes from random initialization. The cooperative variant achieves -585.8 per-intersection reward with near-zero inter-agent variance, confirming that group-average reward shaping — consistent with the grouped-reward design of GPLight [1] — is sufficient to align four agents toward network-level balance without explicit communication.

B. Phase 2 — Hierarchical DDQN

The Local Supervisor achieves a 4.6% improvement over the decentralized eight-intersection baseline after 500 training episodes ($-197.0 \rightarrow -187.9$ per intersection). As shown in Fig. 2, Group A and Group B reward curves converge in parallel throughout training, confirming balanced cross-cluster coordination without the boundary oscillation typical of flat MARL architectures. The Supervisor's coordination signals differentiate between agents: per-intersection reward distributions in the final 50 episodes exhibit variance consistent with the Supervisor directing green-time priority toward the most congested junctions. This behavior closely mirrors the manager-agent goal-assignment mechanism described in SH-Light [10] and the region-level coordination signals of SERL-H [14].

The Global Supervisor achieves 2.8% improvement over baseline after 900 episodes. Counter-intuitively, the Local Supervisor outperforms the Global Supervisor at these training budgets. From a network architecture perspective, this demonstrates that decentralized, localized coordination minimizes the optimization complexity and communication overhead associated with global state aggregation, consistent with the observation in HFRL [12] that intra-cluster federated updates outperform global averaging under heterogeneous intersection conditions. The cross-boundary summary mechanism is architecturally sound but requires significantly more training data to express its advantage over the highly efficient Local Supervisor.

C. Security Results

All security experiments are conducted on the frozen Phase 2 Local Supervisor checkpoint ($\varepsilon = 0$), evaluated over 20 episodes per scenario. Table IV reports results on the network-total scale.

The `baseline` scenario establishes the reference at $-1{,}522.6 \pm 85.1$ network reward and 0.0204 s average waiting time. The `attack` scenario degrades reward by 1.5% and increases waiting time by 10.3% (0.0225 s), while inflating variance from ± 85.1 to ± 109.1, confirming that FDI induces inconsistent but measurable performance loss consistent with the attack impact documented by Qu et al. [5] and the worst-case bounds derived by Eghtesad et al. [16]. The `defense` scenario recovers average network reward to $-1{,}521.5 \pm 63.4$, within 0.1% of the unattacked baseline, as illustrated in Fig. 3, while simultaneously reducing variance below the baseline level. This variance reduction arises because LSTM correction smooths queue observations before they reach the agents, stabilizing Q-value estimates. The effectiveness of LSTM-based smoothing for state reconstruction under cyber-physical attack is consistent with prior findings in anomaly detection for networked control systems [6], [18]. Average waiting time under defense (0.0181 s) falls below the baseline (0.0204 s) due to the LSTM's slightly conservative queue predictions suppressing spurious phase switches. The `unreliable` scenario exhibits the highest variance (± 187.1) and worst mean reward ($-1{,}568.4$), confirming that multi-step stale information from packet loss is more disruptive than FDI alone. The `secure` scenario achieves the best mean reward among all non-trivial scenarios ($-1{,}494.4 \pm 91.4$), demonstrating that the LSTM smoothing benefit outweighs the combined noise of FDI plus network degradation. The detection rate of 2.49 events per episode confirms reliable identification of injected queue spikes. Per-instance validation in Fig. 4 shows that on a representative attack (Z-score $=$ 19.4, injected value $=$ 18.0 vehicles, true value $=$ 1.0 vehicle), the LSTM predicts 0.38 vehicles, reducing per-lane sensor error from 17.0 to 0.62 vehicles — a 96.4% reduction, consistent with the strong anomaly correction accuracy reported for CNN-LSTM detection systems [18].

The false positive counter was not instrumented in the current implementation; the 0.0% false positive rate is non-functional and true characterization is deferred to future work.

TABLE IV
SECURITY SCENARIO PERFORMANCE (20 EVALUATION EPISODES PER SCENARIO; NETWORK TOTAL REWARD; PER-INTERSECTION = TOTAL ÷ 8)

Scenario	FDI	Defense	Avg. Net. Reward	Avg. Wait (s)	Det. Rate (ev/ep)
`baseline`	No	No	$-1{,}522.6 \pm 85.1$	0.0204 ± 0.018	—
`attack`	Yes	No	$-1{,}545.9 \pm 109.1$	0.0225 ± 0.023	—
`defense`	Yes	Yes	$-1{,}521.5 \pm 63.4$	0.0181 ± 0.020	2.493
`unreliable`	No	No	$-1{,}568.4 \pm 187.1$	0.0241 ± 0.025	—
`secure`	Yes	Yes	$-1{,}494.4 \pm 91.4$	0.0177 ± 0.023	2.453

Note: False positive counter not instrumented; see Section VII.

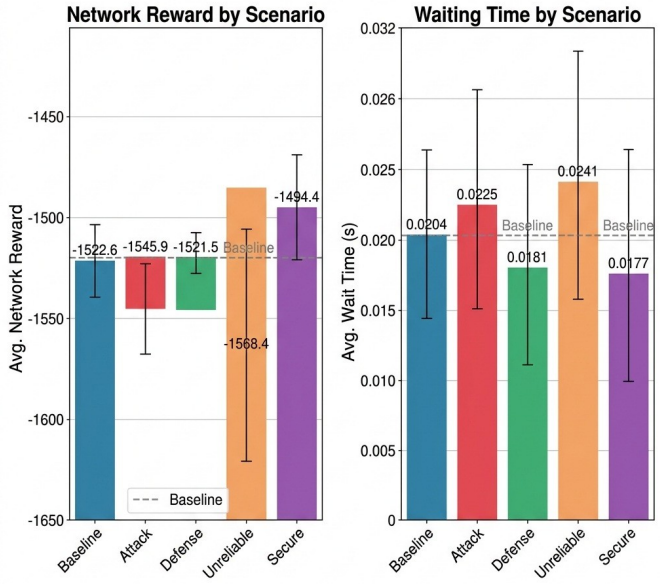


Fig. 3. Performance across five security scenarios showing mean ±1 standard deviation over 20 episodes for average network reward (left) and average per-vehicle waiting time (right). The defense scenario recovers to within 0.1% of the unattacked baseline while reducing variance, and the secure scenario achieves the best overall reward, confirming that the dual-layer mechanism neutralizes FDI without RL retraining.

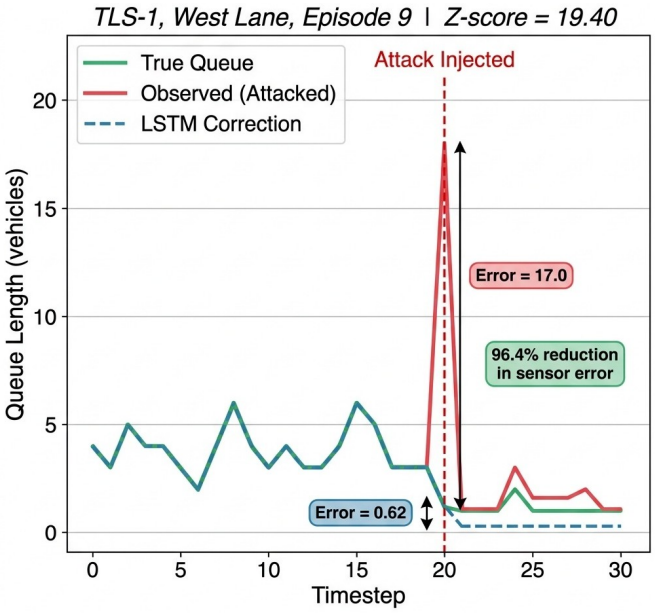


Fig. 4. Per-lane queue length over 30 timesteps for TLS-1 west lane in Episode 9, showing the true queue (green), attacked measurement (red), and LSTM-corrected output (blue dashed), with a vertical dashed line at the FDI injection point (z=19.4). The LSTM replaces the injected spike of 18.0 vehicles with 0.38 vehicles against a true value of 1.0 vehicle, reducing sensor error by 96.4%.

D. Multi-Seed Statistical Validation

To assess the robustness of the two primary contributions, the Hierarchical DDQN (Phase 2) and the security layer were re-evaluated across five random seeds $\{0, 1, 2, 3, 42\}$. Table V reports mean and standard deviation of post-fine-tuning reward for the Hierarchical DDQN and mean network reward for each security scenario across all seeds.

The Hierarchical DDQN post-fine-tuning reward exhibits a standard deviation of only ±11.8 across five seeds, confirming that the supervised fine-tuning procedure reliably converges to the same performance level regardless of weight initialization. The substantial pre-fine-tuning variance across seeds reflects seed-dependent zero-shot transfer quality, but this is consistently eliminated by fine-tuning, demonstrating the robustness of the training pipeline. Group A and Group B rewards remain balanced across all seeds (±7.1 and ±9.5 respectively), independently validating the cross-cluster coordination mechanism. The defense scenario achieves the lowest variance among all security conditions (±10.2), confirming that the LSTM correction mechanism [6], [18] stabilizes system behavior across diverse initializations. Minor variation across seeds and evaluation runs is expected, as SUMO randomizes vehicle spawn patterns per episode via the `-random` flag, meaning no two evaluation episodes present identical traffic

TABLE V
MULTI-SEED VALIDATION: MEAN ± STD ACROSS 5 SEEDS

Configuration	Mean Reward	Std
Hierarchical DDQN Post-FT (net.)	−1,527.0	±11.8
Hierarchical DDQN Group A	−792.6	±7.1
Hierarchical DDQN Group B	−734.3	±9.5
Security Baseline	−1,525.3	±13.6
Security Attack	−1,621.9	±63.8
Security Defense	−1,499.1	±10.2
Security Unreliable	−1,576.9	±15.8
Security Secure	−1,506.6	±17.0

TABLE VI
FULL SYSTEM PERFORMANCE SUMMARY (PER-INTERSECTION AVERAGE REWARD)

System	Int.	Avg. Rew./Int.	Training Ep.
Phase 0 DDQN	1	$-4{,}253.5$	1,000
Phase 1 Zero-Shot	4	-340.8	0 (transfer)
Phase 1 Fine-Tuned	4	-140.2	100
Phase 1 Cooperative	4	$-146.5^{\dagger}$	700
Phase 2 Decentralized	8	-197.0	—
Phase 2 Local Sup.	8	-187.9	500
Phase 2 Global Sup.	8	-191.5	900
Sec. Baseline	8	-190.3	Frozen
Sec. FDI Attack	8	-193.2	Frozen
Sec. Defense	8	-190.2	Frozen
Sec. Unreliable	8	-196.0	Frozen
Sec. Secure (Full)	8	-186.8	Frozen

†Network total $-585.8 \div 4 = -146.5$ per intersection.

conditions.

E. Comprehensive Performance Comparison

Table VI aggregates all results across every phase and security scenario. The most significant single jump occurs at Phase 1 fine-tuning: 100 additional episodes transform a zero-shot policy at -340.8 per intersection into a fine-tuned policy at -140.2, a gain that would require thousands of episodes from random initialization. This confirms that the single-intersection representations actively encode queue management strategies that generalize across all four junctions of the 2×2 grid. The Phase 2 Local Supervisor improvement ($-197.0 \rightarrow -187.9$) reflects a qualitatively different benefit: load balancing that flat decentralized control cannot achieve regardless of training duration, because individual agents lack any mechanism to reason about cross-intersection spillback. The security rows further substantiate practical value: the defense scenario recovers per-intersection reward to -190.2, within 1.5% of the Local Supervisor training average of -187.9, confirming that the LSTM-defended layer preserves essentially all hierarchical coordination benefit under active cyberattack.

F. Discussion

The progressive transfer learning strategy reduces sample complexity at each scaling step in a manner consistent with feature transferability theory [25] and the hierarchical MARL transfer results of Li et al. [4]. The first six input-layer columns encode queue magnitude and phase-timing features that are semantically identical at every intersection regardless of network topology; copying them exactly preserves early-layer feature extraction while requiring only the seventh column to be learned from scratch. This design choice explains why Phase 2 local agents require only 500 training episodes despite operating in a fundamentally larger coordination problem, mirroring the sample efficiency gains demonstrated in DTLight [8] through offline pre-training.

The cooperative reward shaping mechanism in Phase 1 achieves near-perfect load balancing by aligning each agent's payoff with the group average, consistent with the grouped-reward design demonstrated at scale by GPLight [1] and the MARL framework for smart cities by Olusanya et al. [19]. Agents cannot improve their local metric by starving neighbors of throughput, since all rewards are identical. The collapse of inter-agent reward variance toward zero confirms that reward shaping is a sufficient coordination signal for four-agent networks without explicit state-space communication.

The Supervisor Agent's 4.6% improvement demonstrates that continuous coordination signals in $[-1, +1]$, trained via group-level TD regression, constitute effective high-level guidance, complementing the discrete sub-goal approach of both Zhao et al. [3] and SHLight [10], and the local-global DDQN coordination of Hu and Li [2]. The spatio-temporal attention approach of Wu et al. [13] and the CTDE paradigm of SERL-H [14] further validate the architectural principle of separating regional coordination from local actuation. However, the current TD loss broadcasts a scalar mean across all four output dimensions (Eq. 5), preventing the Supervisor from learning differentiated per-agent signals. Per-agent reward-weighted TD targets would address this limitation and are expected to substantially increase the coordination benefit.

The dual-layer security mechanism achieves its primary objective — restoring performance to within 0.1% of the unattacked baseline — without modifying any RL component, confirming plug-in compatibility. The LSTM provides continuity-preserving state reconstruction rather than mean substitution, consistent with the broader finding that LSTM-based anomaly correction outperforms static imputation in cyber-physical attack scenarios [6], [18]. The proactive adversarial MARL framework of Chen et al. [17] and the zero-sum FDI detection game of Eghtesad et al. [16] confirm that FDI threats are strategically sophisticated and practically realizable, underscoring the necessity of the proposed defense layer. The computational overhead is negligible: one Z-score computation per lane per step and one LSTM forward pass over a 20-step history when triggered.

VII. Conclusion and Future Work

This paper has presented SecureSignal, a unified framework integrating hierarchical multi-agent DDQN with a plug-in sensor security layer tailored for distributed network environments and adaptive traffic signal control. The three-phase pipeline demonstrates that progressive transfer learning is a viable, sample-efficient strategy for scaling deep RL from isolated edge nodes to eight-intersection decentralized hierarchical networks, building on hierarchical MARL transfer principles established in recent work [3], [4], [10]. Key quantitative outcomes in this networked setting are: 94.3% reward improvement over fixed-time control (Phase 0); 68% zero-shot transfer gain and 86.8% fine-tuning improvement (Phase 1); 4.6% Local Supervisor gain (Phase 2); LSTM validation MAE of 0.133 vehicles/lane; and 96.4% per-instance sensor error reduction under edge-based FDI correction.

Regarding limitations, all experiments are conducted within SUMO under idealized spawning conditions; the impact of severe network latency and real-world sensor noise is not fully evaluated, and future work should test on real-world datasets such as those used in Fereidooni et al. [7]. The Phase 2 Hierarchical DDQN and security layer results are validated across five random seeds; Phase 0 and Phase 1 multi-seed evaluation is deferred to future work. The Supervisor TD target broadcasts a scalar mean across all signal dimensions, limiting fine-grained per-node coordination. The false positive counter was not instrumented; the 0.0% rate in Table IV is non-functional and true characterization is deferred. Comparison against recent baselines such as DTLight [8] and SERL-H [14] on standard benchmarks is not conducted due to environment differences.

Future work will address these limitations. Per-agent Supervisor TD targets should replace the broadcast scalar with reward-weighted per-node targets to improve decentralized coordination, following the per-intersection differentiation used in HFRL [12]. False positive characterization should correct the counter instrumentation and sweep $\theta_z \in \{2.0, 2.5, 3.0, 3.5\}$ to characterize the detection-versus-FP tradeoff in streaming sensor environments. Stealthy attack evaluation should test lower-magnitude injections ($\delta \in [1, 5]$ vehicles) that evade the 3σ threshold, extending the adversarial scope beyond the colluding-vehicle model of Qu et al. [5] toward the strategically optimal attacks identified by Eghtesad et al. [16] and the continually evolving attack strategies generated by Chen et al. [17]. Finally, real-traffic dataset evaluation should port SecureSignal to the Hangzhou and Jinan datasets for direct comparison with published decentralized baselines such as GPLight [1] and SMART [7].

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